EXECUTIVE BUSINESS REPORT

TOYJOY SEASONAL INVENTORY MANAGEMENT & OPTIMIZATION

A Data-Driven Analysis of Inventory Health, Seasonal Demand & Replenishment Strategy

PREPARED BY

Agamya Jain | Business Analyst

DEPARTMENT

Business Intelligence & Analytics

ANALYSIS PERIOD

April 2023 – March 2026 (3 Years)

REPORT DATE

August 2026 | Portfolio Edition



Strategic Project Objectives

Data-driven strategic initiative designed to resolve warehouse overstock, eliminate high-demand stockouts, and optimize cash flow efficiency across 10 core product lines.

Identify Overstock & Slow-Moving SKUs

Identify products and categories with excess inventory or low sales velocity and assess where inventory levels may be tying up working capital. The analysis will help distinguish healthy inventory levels from products requiring inventory reduction or closer monitoring.

Understand Seasonal & Historical Demand Patterns

Analyze historical sales patterns across months and seasons to identify recurring fluctuations in product demand. These insights will help determine when inventory levels should be adjusted to better align replenishment with expected demand.

Optimize Inventory Allocation & Replenishment

Develop product-level inventory recommendations by balancing demand, stock availability, inventory turnover, and excess-stock risk. The objective is to establish a more effective replenishment approach that reduces both overstock and stockout situations.

Key KPIs Tracked:-

Total Revenue | Excess Inventory Value | Inventory Turnover | Units Sold | Inventory Value | Stockout Rate(%)

Executive Performance Dashboard

Total Revenue
₹32.13M
₹32,132,799 Total sales across 3 years

Excess Inventory Value
₹207.70M
Overstock threshold >2x sales

Avg Inventory Turnover
48.50
Units Sold ÷ Average Inventory

Total Units Sold
53,601
100% data audit completed

Avg Inventory Value
₹39.42K
Average inventory investment

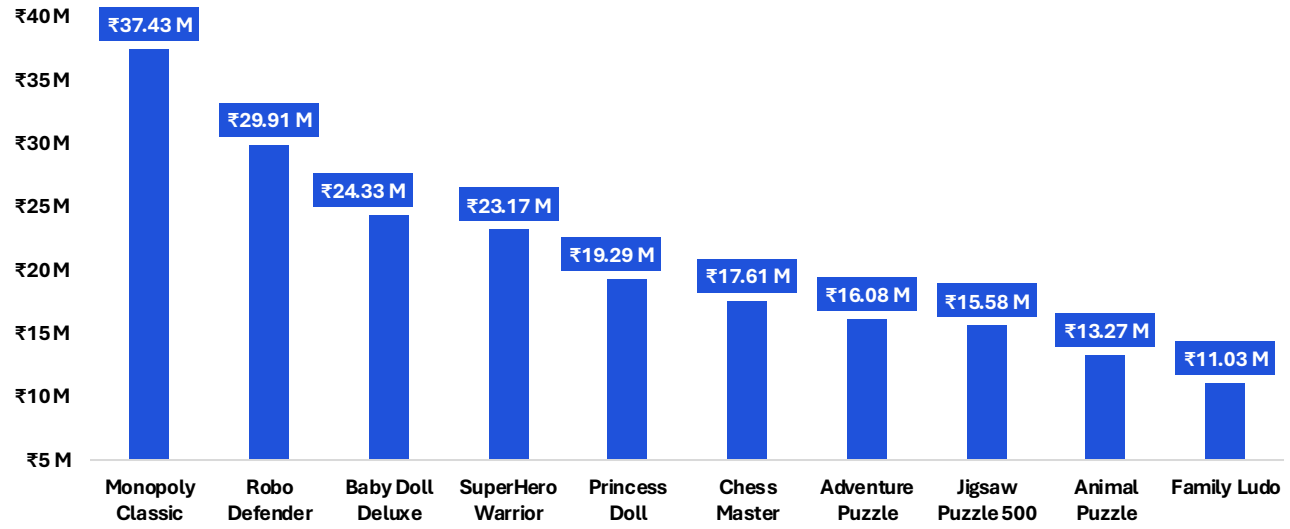
Stockout Rate
0.17%
Low systemic stockout risk

Analysis Flow:-

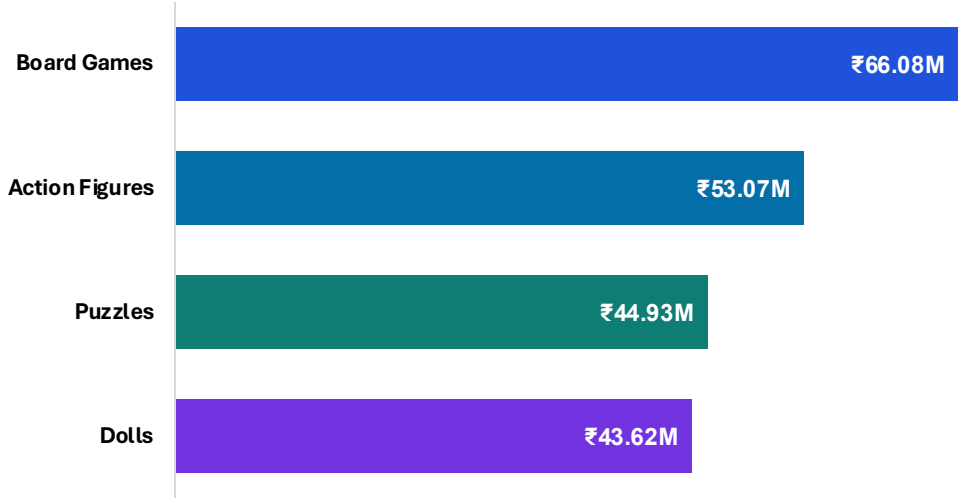
Product Category → Stockout → Seasonal → Inventory Adjustment → Replenishment Strategy → Store Allocation → Recommendation

Q1. Which products or categories have excess inventory that ties up capital?

Top 10 Products by Excess Inventory Value (₹M)



Excess Inventory Value by Category (₹ M)



Capital Concentration

- **Monopoly Classic:** ₹37.43M excess inventory — highest among all SKUs.
- Robo Defender and Baby Doll Deluxe follow at **₹29.91M** and **₹24.33M**.
- **Top 3 SKUs ≈ 44%** of total excess inventory value.

Category Exposure

- **Board Games:** ₹66.08M — highest excess inventory value.
- **Action Figures:** ₹53.07M — second highest.
- **Top 2 categories ≈ 57%** of total excess inventory value.

Management Implications

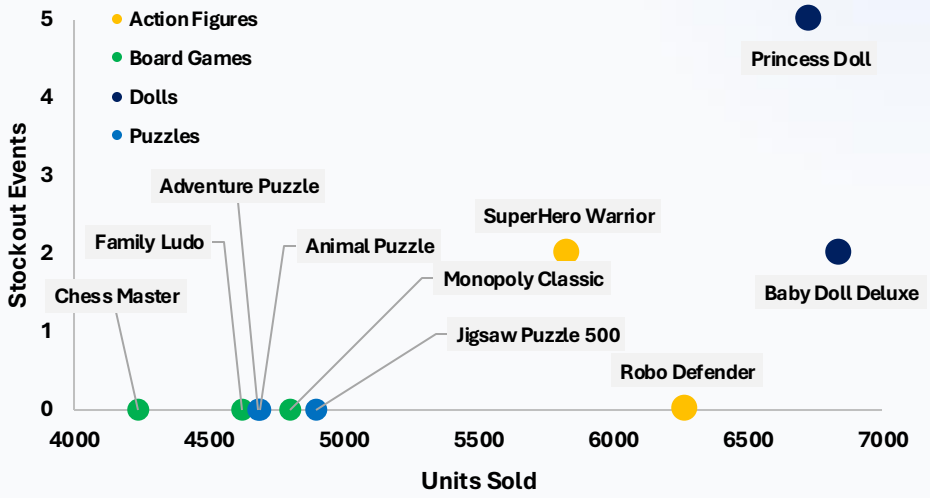
- Prioritize **high-value excess SKUs** for inventory reduction.
- Review replenishment quantities for **Board Games & Action Figures**.
- Focus on **releasing working capital** without creating future stockouts.

Q2. Which products are frequently out of stock, causing lost sales?

Stockout Product Table

Product Name	Category	Stockout Count	Stockout Rate	Inventory Turnover	Total Revenue	Total Units Sold
Princess Doll	Dolls	5 events	0.95%	63.32	₹3.70M	6,731
Baby Doll Deluxe	Dolls	2 events	0.37%	62.71	₹4.44M	6,845
SuperHero Warrior	Action Figures		0.42%	52.49	₹4.08M	5,832
Adventure Puzzle	Puzzles	0 events	0.00%	42.76	₹2.34M	4,680
Animal Puzzle	Puzzles		0.00%	43.28	₹1.87M	4,688
Chess Master	Board Games		0.00%	36.91	₹2.33M	4,237
Family Ludo	Board Games		0.00%	40.48	₹1.38M	4,620
Jigsaw Puzzle 500	Puzzles		0.00%	43.88	₹2.20M	4,895
Monopoly Classic	Board Games		0.00%	42.56	₹4.80M	4,800
Robo Defender	Action Figures		0.00%	56.61	₹5.01M	6,273

Demand vs. Stockout Risk



⚠ Stockout Concentration

- ❑ **3 of the Top 4 sellers** had stockouts.
- ❑ Risk is concentrated in **Dolls & Action Figures**.
- ❑ Indicates **potential lost-sales** risk on high-demand SKUs.

Princess Doll Needs Priority

- ❑ **5 stockout events** — highest observed.
- ❑ **6,731 units sold | 63.32 turnover.**
- ❑ Requires **higher replenishment priority** to reduce potential lost sales.

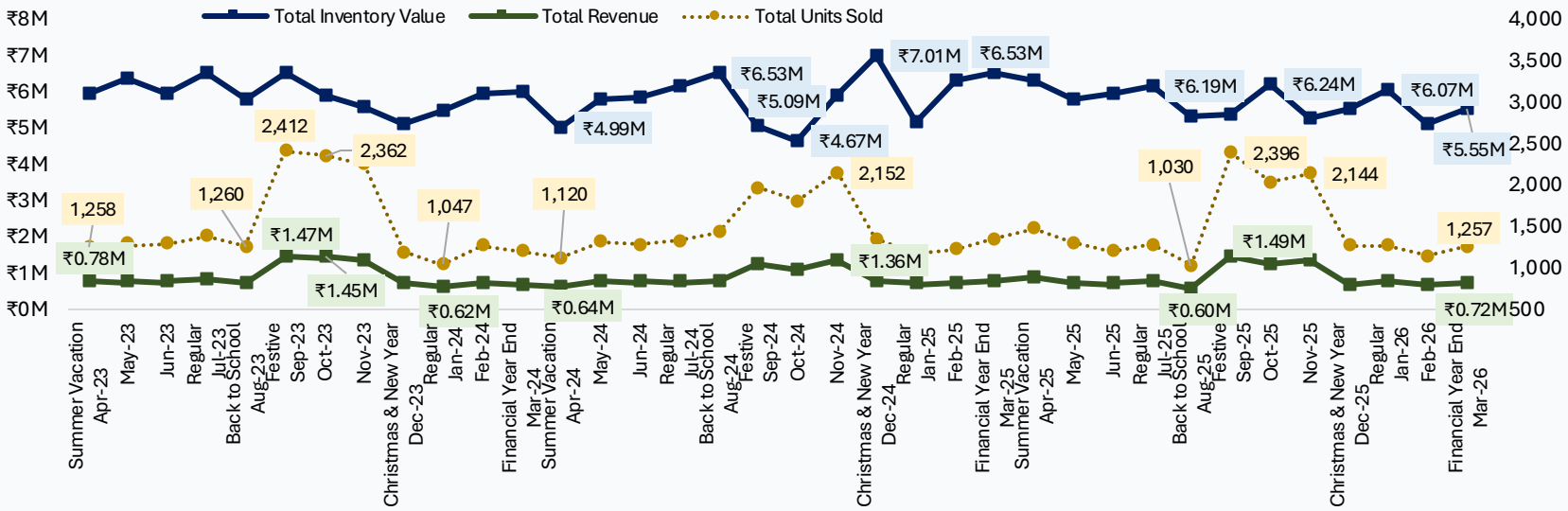
Robo Defender Benchmark

- ❑ **6,273 units sold | 0 stockouts.**
- ❑ High demand can be supported **without stockouts**.
- ❑ Benchmark for **inventory availability.**

EXECUTIVE TAKEAWAY: Prioritize Princess Doll, Baby Doll Deluxe and SuperHero Warrior for higher reorder points and safety stock. Monitor these high-demand SKUs more frequently and replenish proactively before peak-demand periods to **reduce stockouts and minimize potential lost sales**.

Q3. Can you identify seasonal trends in sales to guide stocking strategy?

Seasonal Demand & Monthly Sales Trends (Apr 2023 – Mar 2026)



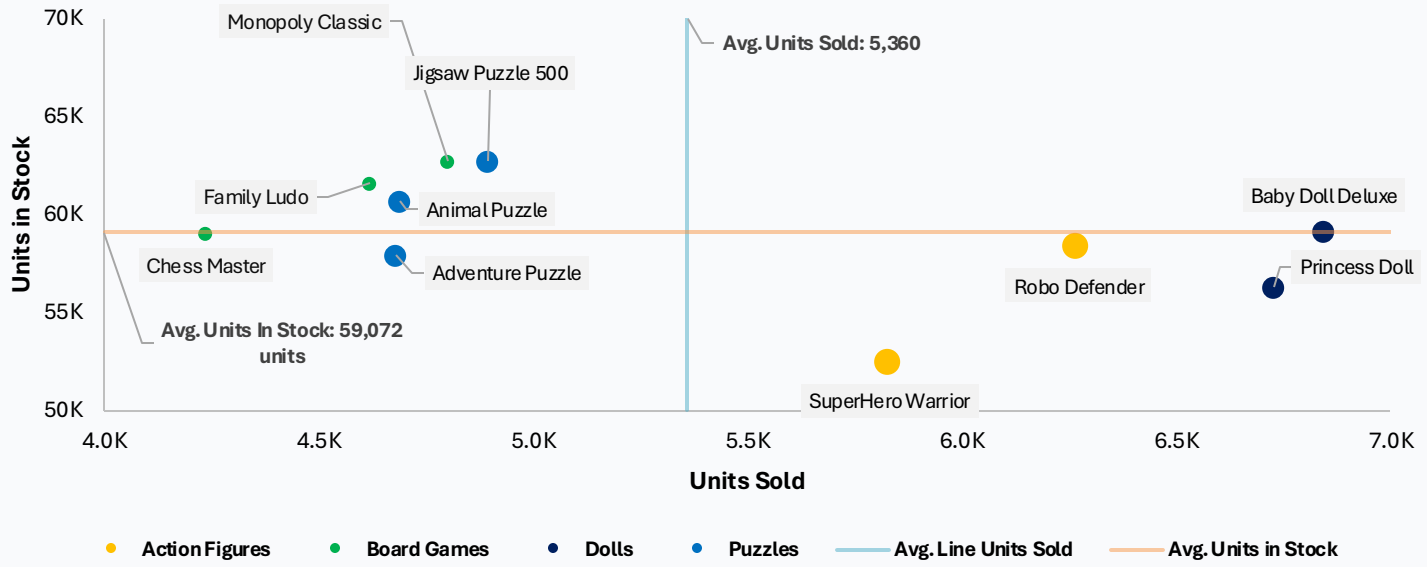
Total Units Sold by Season (3-Year Aggregated)



Peak Season	Secondary Demand	Monthly Peak	Low-Demand Periods	Stock Implications
❑ **Festive:** 19.54K units — highest-demand season. ❑ **~69% higher than Summer Vacation.** ❑ Requires the strongest pre-season inventory buildup.	❑ **Summer Vacation:** 11.56K units. ❑ **Regular Season**: 11.15K units. ❑ Together with Festive, these represent **~79% of total demand.**	❑ **August–September** show recurring demand spikes across the 3-year period. ❑ **2023**: 2,412 - 2,362 units ❑ **2025**: 2,396 - 2,144 units	❑ **Apr, Jun & Jul frequently fall** around the 1.0K–1.3K unit range. ❑ Demand is substantially below the recurring Aug–Sep peaks. ❑ **Inventory levels** can be **normalized during these lower-demand periods.**	❑ **Build inventory before August–September / peak seasons.** ❑ Maintain stronger coverage through high-demand periods. ❑ Gradually normalize stock after demand falls to reduce excess inventory.

Q4. Recommend inventory adjustments for underperforming and high-demand products.

Product Units Sold Vs Units in Stock



Product Name	Total Units Sold	Total Inventory	Total Revenue	Inventory Turnover
Baby Doll Deluxe	6.85K	59.16K	₹4.44M	62.71
Princess Doll	6.73K	56.24K	₹3.70M	63.32
Robo Defender	6.27K	58.29K	₹5.01M	56.61
SuperHero Warrior	5.83K	52.45K	₹4.08M	52.49
Jigsaw Puzzle 500	4.90K	62.69K	₹2.20M	43.88
Monopoly Classic	4.80K	62.71K	₹4.80M	42.56
Animal Puzzle	4.69K	60.66K	₹1.87M	43.28
Adventure Puzzle	4.68K	57.90K	₹2.34M	42.76
Family Ludo	4.62K	61.64K	₹1.38M	40.48
Chess Master	4.24K	59.00K	₹2.33M	36.91

High-demand & Fast-moving

- Baby Doll Deluxe: 6.85K units | ₹4.44M | 62.71× turnover
- Princess Doll: 6.73K | ₹3.70M | 63.32×
- Action: Maintain strong allocation for high-demand, efficiently moving products.

Lower-demand & Slower-moving

- Chess Master: 4.24K units | ₹2.33M | 36.91× turnover
- Family Ludo: 4.62K | ₹1.38M | 40.48×
- Action: Reduce future allocation for lower-demand, slower-moving products.

Prioritize By Demand + Efficiency

- Robo Defender: 6.27K units | ₹5.01M | 56.61×
- SuperHero Warrior: 5.83K | ₹4.08M | 52.49×
- **Action:** Protect allocation for commercially important products with strong demand and turnover.

Q5. Suggest strategies to optimize reorder points and quantities to reduce overstock and stockouts.

Product Name	Inventory Turnover	Stockout Count	Excess Inventory Value	Recommended Action
Princess Doll	**63.32**	**5 events**	₹19.29M	Increase ROP & safety stock; replenish more frequently
Baby Doll Deluxe	**62.71**	**2 events**	₹24.33M	Increase ROP selectively; monitor excess stock closely
Robo Defender	56.61	0	₹29.91M	Maintain ROP; reduce order quantity to control excess
SuperHero Warrior	52.49	**2 events**	₹23.17M	Increase ROP moderately; replenish more frequently
Jigsaw Puzzle 500	43.88	0	₹15.58M	Maintain lower ROP; reduce reorder quantity
Animal Puzzle	43.28	0	₹13.27M	Maintain lower ROP; avoid large replenishment batches
Adventure Puzzle	42.76	0	₹16.08M	Reduce reorder quantity; replenish conservatively
Monopoly Classic	42.56	0	₹37.43M	Lower reorder quantity; tightly control replenishment
Family Ludo	40.48	0	₹11.03M	Lower ROP & reorder quantity; replenish conservatively
Chess Master	36.91	0	₹17.61M	Lowest ROP & reorder quantity; minimize replenishment

Increase Replenishment

High Turnover + Recurring Stockouts

Princess Doll | Baby Doll Deluxe| SuperHero Warrior

↓

Higher reorder point + more frequent replenishment

Reduce Reorder Quantity

Lower Inventory Turnover

Chess Master | Family Ludo

↓

Lower purchase quantities + monitor excess

Controlled Replenishment

High Excess Inventory + Moderate Turnover

Monopoly Classic

↓

Avoid aggressive replenishment; review inventory before reordering.

Fast-moving + Stockout Risk

Princess Doll: 63.32 turnover + 5 stockouts.
Baby Doll Deluxe: 62.71 + 2 stockouts.
SuperHero Warrior: 52.49 + 2 stockouts.
Action: Increase reorder priority and replenish more frequently.

Slower Inventory Movement

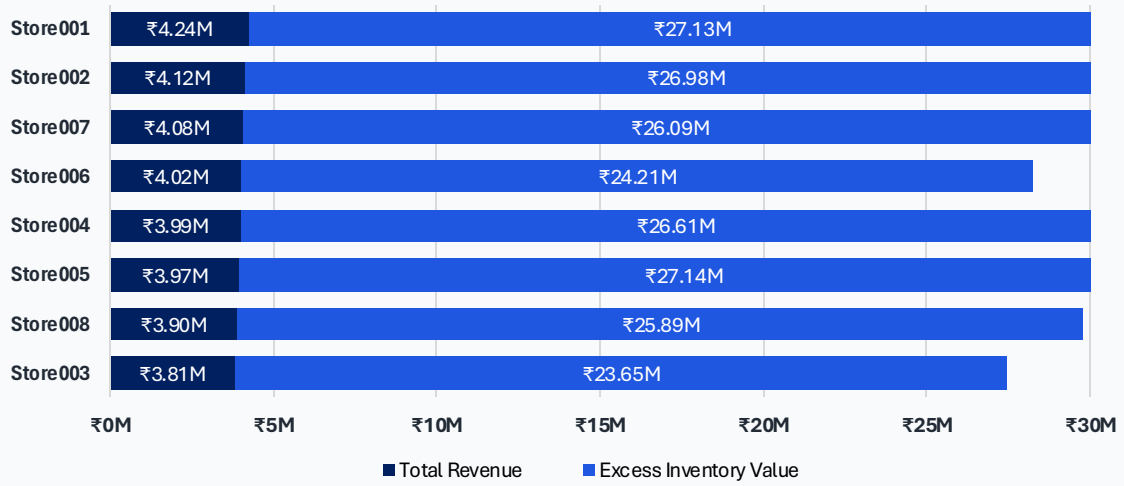
Chess Master: lowest turnover at **36.91**.
Family Ludo: 40.48 turnover.
Action: Reduce future reorder quantities to limit excess inventory.

High-value Excess Risk

Monopoly Classic: ₹37.43M excess inventory. Turnover is only **42.56**, below the portfolio high of 63.32.
Action: Maintain controlled replenishment and closely monitor stock before reordering.

Store Performance & Inventory Allocation Analysis

Revenue & Excess Inventory Value by Store ID (₹ M)



Store ID	Net Revenue	Total Units	Turnover Rate	Stock Outs	Top 3 Product Volume Drivers (Units Sold)
Store001	₹4.24 M	6,969	61.60x	3	Baby Doll Deluxe (1,148) | SuperHero Warrior (895) | Princess Doll (859)
Store002	₹4.12 M	6,887	59.85x	1	Princess Doll (1,044) | Robo Defender (1,008) | Baby Doll (754)
Store006	₹4.02 M	6,657	63.45x	0	Robo Defender (1,065) | Baby Doll (966) | SuperHero (757)
Store007	₹4.08 M	6,722	60.12x	1	Baby Doll Deluxe (950) | Princess Doll (866) | Robo Defender (746)
Store004	₹3.99 M	6,645	60.01x	0	Baby Doll Deluxe (777) | SuperHero (769) | Princess Doll & Jigsaw Puzzle 500 (759)
Store005	₹3.97 M	6,704	59.20x	2	Princess Doll (925) | Baby Doll (895) | SuperHero (739)
Store008	₹3.90 M	6,629	59.08x	2	Robo Defender (847) | Adventure Puzzle (761) | Jigsaw 500 (731)
Store003	₹3.81M	6,388	60.81x	0	Princess Doll (868) | Baby Doll (775) | SuperHero (747)

High-Volume Store Bottlenecks

- ❑ **Store001 & Store002**
- ❑ Contribute **26.01% of revenue (₹8.36M)**
- ❑ **44.4%** of **stockouts (4/9)**

→ Prioritize replenishment review.

Store006 : Efficiency Benchmark

- ❑ **Highest inventory turnover (63.45×)**
- ❑ No recorded stockouts
- ❑ **Lowest inventory holding (68.4K units)**

→ Benchmark for efficient replenishment.

Store008 — Puzzle Demand

- ❑ **Highest volume of Puzzles sold.**
- ❑ **Adventure Puzzle (761), Jigsaw Puzzle 500 (731)** and **Animal Puzzle (685)**

→ Allocate more puzzle inventory to Store008.

EXECUTIVE TAKEAWAY: Replace uniform chain-wide replenishment with store-cluster allocation. Prioritize puzzle inventory allocation toward Store008 based on higher observed puzzle demand, while **increasing replenishment frequency for star Doll and Action Figure SKUs at Store001 and Store002**.

Strategic Recommendations & Action Plan

QUICK WINS (0–3 Months)

- **Rebalance Overstock:** Execute promotional bundling for Monopoly Classic (₹37.43M excess) to free up warehouse space.
- **Elevate Princess Doll Buffer:** Increase safety-stock coverage for Princess Doll and monitor stockout recurrence to prevent repeat stockouts.
- **Festive Pre-build:** Initiate festive procurement 6 weeks in advance (Aug/Sep) to capture the 19.5K unit seasonal surge.
- **Slow-Mover Reorder Freeze:** Reduce reorder quantities for Chess Master & Family Ludo and monitor excess inventory.

LONG-TERM STRATEGIES (3–12 Months)

- **Power BI Automated Forecasting:** Implement real-time Power BI inventory alerts linked with store transactions.
- **Supplier Lead-Time Audit:** Partner with manufacturers to reduce procurement lead times for Star SKUs.
- **Warehouse Optimization:** Reallocate shelf capacity from Board Games toward high-turnover Dolls & Action Figures.
- **Stimulate Off-Peak Demand:** Test gifting bundles and back-to-school kits during weaker periods—Back to School (3.72K), Financial Year End (3.81K), and Christmas/New Year (3.82K)—to increase sales during these periods.

Thank You | Questions & Discussion

Prepared by Agamya Jain | Data Analyst | Business Intelligence & Analytics Department
ToyJoy